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Effect of joint gap on bead formation in laser butt welding of stainless steel

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ABSTRACT

Simulations and experiments of laser butt welding are presented in the paper with the laser beam traveling along an inclined line across the gap. Full penetration was observed when the laser beam focused on the gap, while partial penetration was achieved when the laser beam deviated from the gap. A three-dimensional numerical model coupled with a ray tracing algorithm was established to investigate and compare the transient dynamics of the keyhole, molten pool and laser induced plume. The shape of the simulation cross-sections of the welds show good agreement with the experimental results. The gap contributed to transmission of laser beam energy deep into the inner part of the workpiece. The absorbed energy by the workpiece decreased when the laser beam was focused on the gap as some rays escaped through the opening gap. The decrease in absorbed energy resulted in a lower velocity of the molten pool and lower average pressure on the free surface. The velocity and volume of the plume on the top side of the workpiece also decreased. Flow patterns in the molten pool were also identified and discussed.

1. Introduction

Butt welding technology of two thick workpieces is important in shipbuilding, pipeline, nuclear, and submarine manufacturing field (Li et al., 2014). Many problems, such as residual stress and residual deformation (Hu and Zhang, 2001), arise because of the large groove area that is necessary to join thick pieces with traditional methods. Many researchers have carried out studies on high-power fiber laser butt welding of thick workpiece due to its high efficiency, small welding deformations, easy installation, small focused beam and a robust setup for mobile applications.

Lee et al. (2010) applied a Gaussian surface distribution of heat source combined with electric arc or the laser beam to investigate the effect of temperature field on sensitization of Alloy 690 butt welds. Chen et al. (2014a) studied the metallurgical morphology and strength of weld joint during laser butt welding of 4 and 2 mm SUS301L stainless steel. Chen et al. (2014b) studied the microstructures and mechanical property of laser butt welding of titanium alloy to stainless steel. Atabaki et al. (2016) experimentally applied autogenous laser welding and laser welding assisted with a cold wire to join thick grooved plates of structural steel (A36) in a horizontal narrow gap butt joint configuration. They evaluated the weld bead shape, cross-section, and mechanical properties. Gou et al. (2016) studied properties of the welded joint of single-pass fiber laser butt welding of explosively welded 2205/X65 bimetallic sheets. Although, the studies mensioned above mainly focus on the microstructure and strength of the laser butt welding bead, the transmission of laser ray energy and the dynamics of the keyhole, molten pool, and plume were not involved.

Laser welding is a multi-phase, multi-physics process (Dal and Fabbro, 2016). Due to the high-intensity irradiation energy of laser beam in deep penetration laser welding, the workpiece material melts and subsequently evaporates. Evaporation will generate recoil pressure on the free surface (Cho and Na, 2006) and a Marangoni flow is generated on the keyhole wall due to the high gradient of the surface temperature. The recoil pressure and the Marangoni flow force drive the flow of liquid material outwards and hence a keyhole is formed. The laser rays were multi-reflected in the formed keyhole, and consequently the beam energy is transmitted into the inner part of the workpiece. The keyhole is maintained by the pressure and energy balance on the free surface. Plume induced by metal vapor attenuates the energy of laser beam due to its scattering and absorption effect of the laser rays. In general, a better understanding of dynamics of the keyhole, molten pool and plume is crucial for the deep penetration laser butt welding.

Experimental study of the concomitant dynamics of the keyhole, molten pool and plume with satisfactory details is difficult. Visual-based observations coupled with CCD cameras have been carried out. Gao and Sun (2014) developed a visual system to monitor keyhole dynamic characteristics during high power disk laser welding of Type

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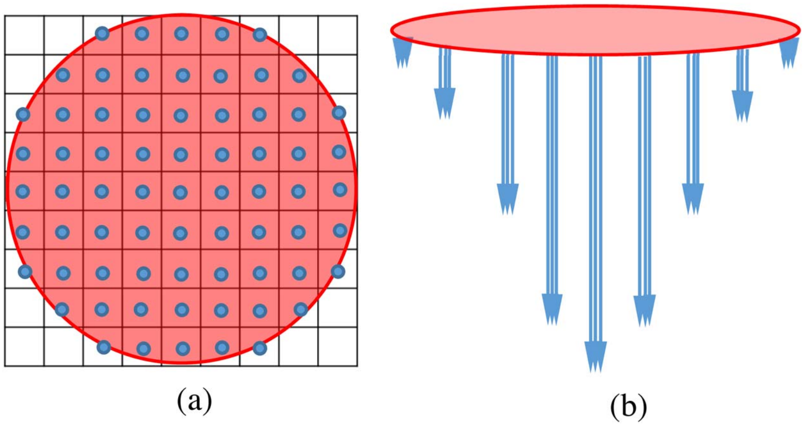

Fig. 1. Modeling of the Gaussian distribution laser beam. (a) Sub-nodes on the focal plane; (b) Gaussian distribution of laser beam energy.

304 austenitic stainless steel. Zhang et al. (2015) applied a high-speed camera to capture the dynamics of the molten pool during high power disk laser welding. Kawahito et al. (2011) built an x-ray transmission imaging system to observe the keyhole and molten pool for in-situ analysis during 10 kW fiber laser welding. They provided an improvement of understanding of keyhole evolution and flow-pattern of the molten pool. However, the application of X-ray transmission for in-situ observation is limited due to its high cost and possible harm caused by X-ray radiation. Furthermore, the interaction of laser beam and the material in the inner part of the workpiece is difficult to be observed. The laser-material interaction in this part of the workpiece is crucial since it highly affects the distribution of the laser energy on the keyhole wall.

Many numerical simulation models, including self-consistent keyhole formation, were developed to gain a further understanding of deep penetration laser welding process. The geometry of the free surface highly affects the propagation of laser rays. Volume-of-Fluid (VOF) is a powerful method of reconstructing the geometry of free surface in numerical simulation. Researchers applied VOF-based numerical simulation models to investigate the temperature field (Wu et al., 2009), keyhole behavior (Zhao et al., 2011), molten pool behavior (Amara and Fabbro, 2008), porosity formation (Zhao and DebRoy, 2003), humps formation (Amara and Fabbro, 2010) and transport phenomena (Zhou et al., 2006) during laser welding. The research group of Suck-Joo Na developed a series of three-dimensional transient models of deep penetration laser welding based on VOF coupled with a ray tracing algorithm. The established models were used to investigate the laser energy deposition on the free surface (Cho and Na, 2006), and were also extended to laser-arc hybrid welding simulation (Cho et al., 2010). Ahn and Na (2013) also developed a new free surface reconstruction method by the combination of the VOF and Level-set method. This new method produced a smoother and continuous free surface. Although, these studies are mainly focusing on Bead-on-Plate (BOP) laser welding, the behaviors of the keyhole, molten pool and plume during laser butt welding were not involved.

In this work, a three-dimensional numerical model was established to investigate the effect of the joint gap on the transmission of laser beam energy and the dynamics of the keyhole, molten pool, and plume during laser butt welding. Buoyancy force, Marangoni force, recoil pressure, the vapor-induced plume heat transfer and plume velocity were modeled in the simulation. The transmission of laser beam energy and the behaviors of molten pool, keyhole, and plume in laser butt welding and BOP welding were also compared and analyzed.

2. Mathematical models

2.1. Assumptions and governing equations

In this work, the following assumptions were employed.

(1) The molten metal is assumed to be an incompressible laminar flow with Newtonian viscosity.
(2) The region of gas or plume is treated as a void because solid or liquid steel has a greater density level than gas or plume. Gas bubbles in the molten pool are treated as adiabatic bubbles.
(3) The energy density of the laser beam is assumed to be a Gaussian distribution, and the recoil pressure induced by the evaporation is determined by Clausius-Clapeyron equation.
(4) The metallic vapor generated and ejected through the keyhole entrance in laser welding is modeled as an ideal gas and assumed to be an extra heat source during welding with a temperature of 6000 K (Cho et al., 2012).
(5) Inside the blind keyhole, the velocity of vapor plume is assumed to decrease linearly along the keyhole depth from the calculated velocity at the entrance of the keyhole to zero at the bottom of the keyhole. If full penetration is achieved, the keyhole is open at the bottom. The velocity of the vapor plume in the upper half of the keyhole is assumed to decrease linearly from the calculated velocity at keyhole entrance to zero at the half thickness point (the vapor plume flows upward). The vapor plume in the lower half of the workpiece flows downward, and its velocity is assumed to linearly increase from zero at the half thickness to the calculated velocity at the exit of the keyhole in the bottom of the workpiece.

The governing equations of mass conservation, momentum conservation (Navier-stokes), energy conservation and VOF equation were solved to analyze the heat transfer and fluid flow during the laser welding process. A detailed introduction of these equations can be found in previous work of Na' group (Cho and Na, 2006; Cho et al., 2010, 2012).

2.2. Physical models in simulation

The Laser beam is modeled by a geometrical optics method and defined as a bundle of laser rays. The focal plane is divided into a set of sub-nodes shown in Fig. 1(a). The laser beam energy is assigned to each sub-node depending on the distance between the sub-node and the focal center based on Gaussian distribution function in Eq. (1).

$$q_{\mathrm{L}}(x, y, z) = 2P_{\mathrm{L}}/\pi r_{\mathrm{f0}}^2 \exp(-2(x^2 + y^2)/r_{\mathrm{f0}}^2) \tag{1}$$

Here, rf0 is the focal radius, PL is the laser power. Each sub-node inside the focal plane contains a certain number of sub rays shown in Fig. 1(b). These sub rays have the same energy, with sum of energy equal to the energy assigned to this sub-node. The initial direction of the laser rays is calculated in a diffractive manner, and the beam radius at position z along the axis of the beam is expressed in Eq. (2).

$$r_f(z) = r_{\mathrm{f0}}[1 + ((z - z_0)/z_r)^2]^{1/2} \tag{2}$$

where z0 is z position at focal length plane while zr is Rayleigh length, and zr = ± 2rfoF.

The laser absorptivity on the workpiece can be determined by experimental properties of the material and the Fresnel reflection model. A detailed description of the calculation method of vapor velocity and the calculation of heat impact of the vapor on the keyhole wall, together with a description of the model of laser beam absorption, recoil pressure, Marangoni flow, Buoyancy force, can be found in Cho's paper (Cho et al., 2012).

The algorithm of free surface tracking in this research is based on a combination of VOF and the Level-set method. A detailed introduction of interface tracking and ray tracing algorithm is given by Ahn's paper (Ahn and Na, 2013).

2.3. Boundary condition

The beam energy on the top surface is balanced by the laser heat flux, the heat dissipation by convection and radiation, and the heat loss due to the surface evaporation. The energy balance is expressed by Eq. (3).

$$k\frac{\partial T}{\partial \vec{n}} = q_{\mathrm{L}} - q_{\mathrm{conv}} - q_{\mathrm{rad}} - q_{\mathrm{evap}} = q_{\mathrm{L}} - h(T - T_a) - \varepsilon_r\sigma(T^4 - T_a{}^4) - q_{\mathrm{evap}} \quad (3)$$

where $\vec{n}$ is the vector normal to the local free surface, h is the convective heat transfer coefficient, Ta is the ambient temperature, εr is the surface radiation emissivity, and σ is the Stefan-Boltzmann constant. The variation of energy and mass by evaporation is calculated by considering the evaporated mass flux under the assumption that roughly 18% of all evaporating plumes are re-condensed on the surface of the workpiece (Allmen and Blatter, 1995).

The gap surface is also considered as the free surface. When the laser beam was focusing on the gap area, the laser ray would contact the gap surface, and its energy was absorbed. Therefore, the energy on these two gap surfaces can be expressed by Eq. (3). The energy on the bottom free surface is expressed in a similar manner in Eq. (3), but the laser heat source should be eliminated if the keyhole is blind.

The pressure boundary condition is expressed by Eq. (4).

$$p = p_{\mathrm{R}} + \frac{\gamma}{R_{\mathrm{c}}} \quad (4)$$

here, γ is the surface tension coefficient and Rc is the radius of the surface curvature.

3. Experimental and simulation setup

3.1. Experimental setup and results

Type 304 stainless steel with a thickness of 10 mm is used in both simulations and experiments. The parameters during the laser butt welding are tabulated in Table 1. A schematic of the experimental setup is shown in Fig. 2.

Table 1
The parameters of the welding experiment.

Welding parameters	Values
Laser power	6 kW
Beam diameter	200 μm
Focus position	0 mm
Welding speed	2.5 m/min
Welding materials	Stainless steel (Type 304)
Welding material dimensions (L × W × D mm)	150 × 49 × 10
Gap size	0.1 mm
Deviations between beam and gap centerline	From 1 mm gradually to (−1) mm

Fig. 3 displays the experimental results. The top and bottom view of the workpiece demonstrates that full penetration is achieved in the center area, indicated by the dashed border rectangle. Nevertheless, partial penetration was produced in front of and after this center area. When the laser beam focuses on the gap, laser butt welding occurs, and when the laser beam deviates from the gap for a certain distance, this can be categorized as BOP laser welding. The experimental result shows the differences weld bead between the two types of laser welding. Fig. 3 also shows in the center area where full penetration occurred, the width and height of weld bead are smaller than other regions with partial penetration.

3.2. Simulation setup

Four different simulations were designed and implemented to investigate the different mechanism between laser butt welding and BOP laser welding. The schemes are shown in Fig. 4. In Fig. 4 (a), the laser beam travels across the gap. The deviation of the starting point and ending point between the laser beam and gap is 1 mm. In Fig. 4(b), the laser beam track is parallel with the gap, and the deviation is 1 mm. In this situation, the keyhole is far away from the gap, and the process behaves as BOP laser welding. The third scheme in Fig. 4(c) shows that the laser beam track is parallel with the gap with a deviation of 0.5 mm. Fig. 4(d) illustrates the simulation when the laser beam is focused on the gap.

Fig. 5 shows a schematic of the simulation domain. The centered domain (marked in red) is equally divided into fine cells, and the outer domains are divided gradually into coarse cells. The size of the uniform fine cell is 0.05 mm. Due to the computational cost, the simulation times was limited to 100 ms and consequently the traveling distance of laser beam is 4.1667 mm. The width of the gap is set to 0.1 mm in all simulations and experimental trials.

The thermal properties of type 304 stainless steel (Tan et al., 2013; Rai et al., 2009) are shown in Table 2. The thermal conductivity of the solid material is temperature-dependent, and thermal conductivity of the liquid material is set as a constant due to the inadequacy of the high-temperature properties of the material.

3.3. Results comparison and validation

The simulated weld bead cross sections are compared with the experimental results to verify the laser butt welding model and algorithms in this work. The cross sections of experimental weld beads were polished and etched to show the fusion zone. Fig. 6(a) shows the fusion zone when the deviation between laser beam and gap was 0.5 mm. For this trial, partial penetration was observed and the molten pool depth at this stage was about 8.75 mm. Fig. 6(b) shows the fusion zone when the laser beam was focused on the gap, and full penetration is observed. The two simulation results in Fig. 6 were acquired from simulation Case 1. The profile of the experimental fusion zone is overlaid on the simulated weld bead. The comparison shows that the simulation results are in good agreement with the experimental results except for the top surface. The top surfaces of the experimental results are convex, while the calculated results have a concave or flat top surface. It is known that 8–12% of the material volume is increased due to the thermal expansion in deep penetration laser welding process (Sudnik et al., 2000) and thus the observed convex top surface occurs as shown in Fig. 6. In this work, the increase of material volume by its thermal expansion was not considered in the simulation, this could explain the concave or flat top surface. Secondly, mass loss induced by the spatters and evaporation could lead to the concave or flat top surface observed in the simulation results.

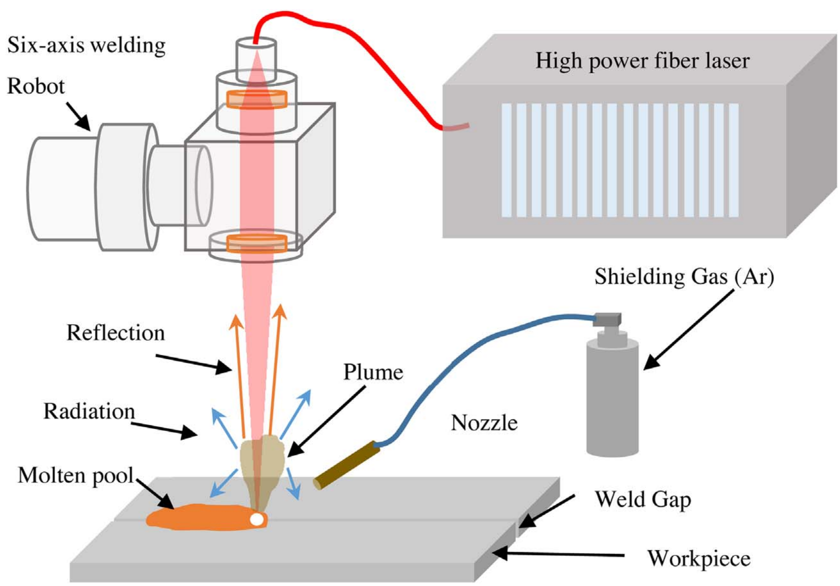

Fig. 2. The sketch of the setup of the experiment.

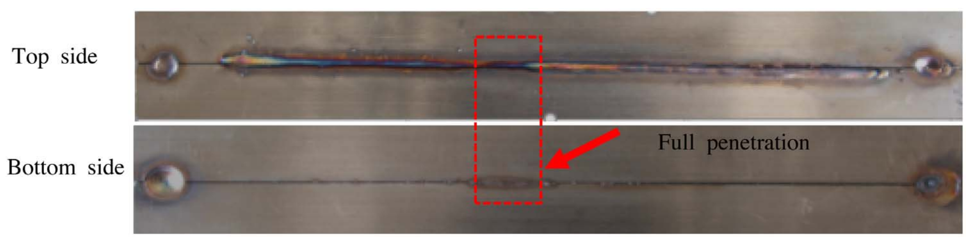

Fig. 3. Top and bottom view of the experimental results.

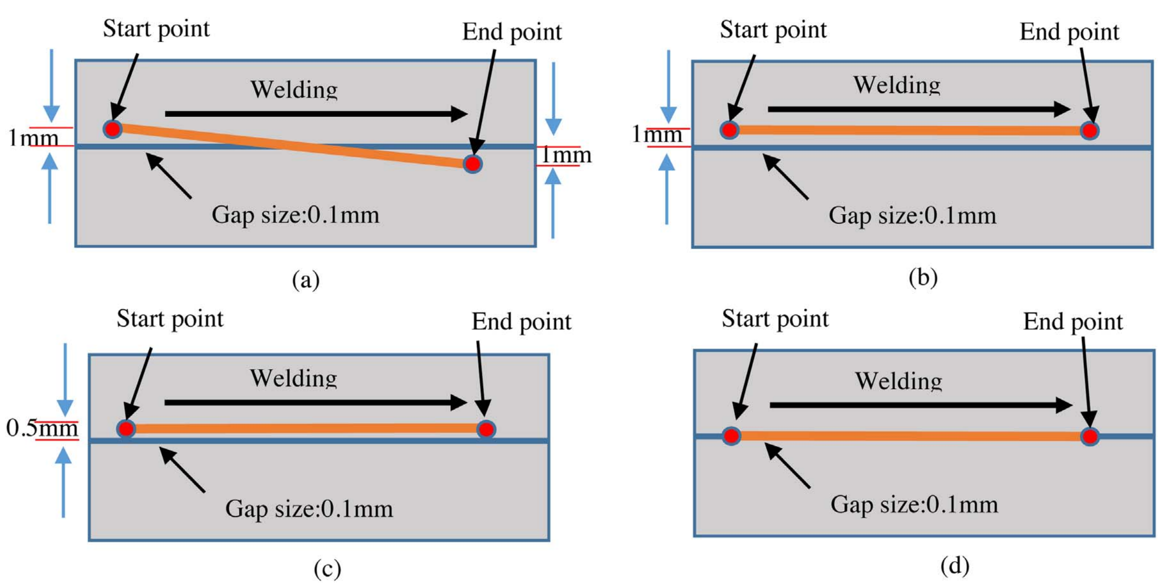

Fig. 4. The four simulation schemes in this work. (a) Case 1-Laser beam crossed gap: Deviation from 1 mm to −1 mm; (b) Case 2-Deviation between laser beam and gap: 1 mm; (c) Case 3- Deviation between laser beam and gap: 0.5 mm; (d) Case 4-Laser beam focused on gap: 0.5 mm.

4. Discussion

4.1. Keyhole and molten pool formation

Fig. 7(a) and (b) shows the formation of the keyhole and molten pool in simulation when the laser beam is focused on the gap (Case 1), and when the laser beam deviated from the gap and focused on the workpiece (Case 1), respectively. For Case 1 it is observed that the laser beam energy quickly transmitted deep into inner part of the workpiece at stage $t1$, $t2$, and $t3$. At stage $t4$, $t5$ and $t6$, the molten pool and keyhole became deeper, with the keyhole width slowly increasing. However, for Case 2 in Fig. 7(b), the depth of the molten pool and keyhole increased slowly, and the width increased quickly with the molten pool driven to the top part of the workpiece. In Fig. 7(a), the penetration increased at stage $t7$, and full penetration was observed at stage $t8$. In Fig. 7(b), the depth of molten pool increased slowly, and only partial penetration was observed. The volume of liquid in the molten pool was larger, and the temperature of the molten pool was higher for Case 1 when compared to Case 2.

These phenomena can be analyzed as follows. When the laser

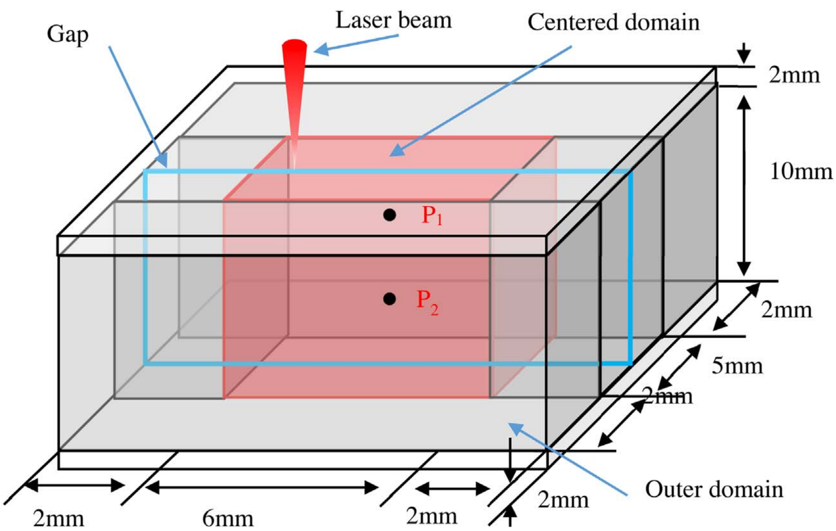

Fig. 5. The computation area setup of the simulation.

Table 2
The thermal properties of the 304-stainless steel.

Properties	Values
Solid density	7900 kg/m^3
Liquid density	6900 kg/m^3
Viscosity	5.9×10^{-3} kg/m s
Surface tension	1.87 N/m
Surface tension gradient	-0.43×10^{-3} N/m K
Specific heat of solid	760 J/kg K
Specific heat of liquid	720 J/kg K
Latent heat of fusion	2.47×10^5 J/kg
Latent heat of vaporization	6.52×10^5 J/kg
Liquids temperature	1697 K
Solidus temperature	1727 K
Boiling temperature	3023 K

beam focused on the gap, the propagation direction of the laser beam was parallel with the gap direction, and the incident angles of laser ray, close to the center of the Gaussian distribution of the laser beam, ranged from 85° to 90°. Fig. 8(a) shows the beam propagation inside the gap when the laser rays contact the gap surface of the workpiece. These laser rays carried a significant portion of the energy of the laser beam. The reflectivity of the workpiece which depends on the incident angle φ and laser-and-material-dependent variable ε, can be calculated based on Eq. (5) with $\varepsilon = 0.33$ ($\varepsilon 2 \approx 1/n22$, n2 is the refractive index of the material, here n2 = 2.9565). In Fig. 8(b) the reflectivity dependence is shown; the beam reflectivity indicates that a large portion of the laser ray energy was reflected when the incident angles ranged from 85° to 90°.

$$R = \frac{1}{2}\left(\frac{1 + (1 - \varepsilon\cos\varphi)^2}{1 + (1 + \varepsilon\cos\varphi)} + \frac{\varepsilon^2 - 2\varepsilon\cos\varphi + 2\cos^2\varphi}{\varepsilon^2 + 2\varepsilon\cos\varphi + 2\cos^2\varphi}\right) \tag{5}$$

Fig. 8(a) also shows that the polished gap surface acted like a mirror at the starting stage: the laser rays were reflected with the same incident angle in the following contacts until the energy of the laser ray was totally absorbed or the laser ray escaped through the gap. In this way, the energy of the laser beam was transmitted through the deep inner part of the workpiece.

In Case 2 shown in Fig. 7(b), the formation and maintenance of the keyhole mainly depended on the recoil pressure and thermocapillary force, but the surface tension in the molten pool drove the fluid flow towards the axis of the keyhole and attempted to close the keyhole. The combination of these forces made the keyhole wall much rougher than the keyhole wall in Fig. 7(a), and this would intensify the absorption of the laser rays' energy in a local area. Consequently, an incresed amount of material near the keyhole wall was melted and the width of molten pool increased compared to Case 1 in Fig. 7(a). This process can be validated in Fig. 7(b) from the stages $t3$ to $t8$; the width of the molten pool increased quickly, but the penetration increased slowly. Moreover, when an equilibrium of these forces was achieved, the width of the molten pool and the penetration kept steady, until the welding process finished.The variations of keyhole depth and radius in the four simulations were investigated and shown in Fig. 9. The variation of keyhole radius and keyhole depth kept relatively steady after $t = 30$ ms, which meant that the laser welding process was steady from then on.

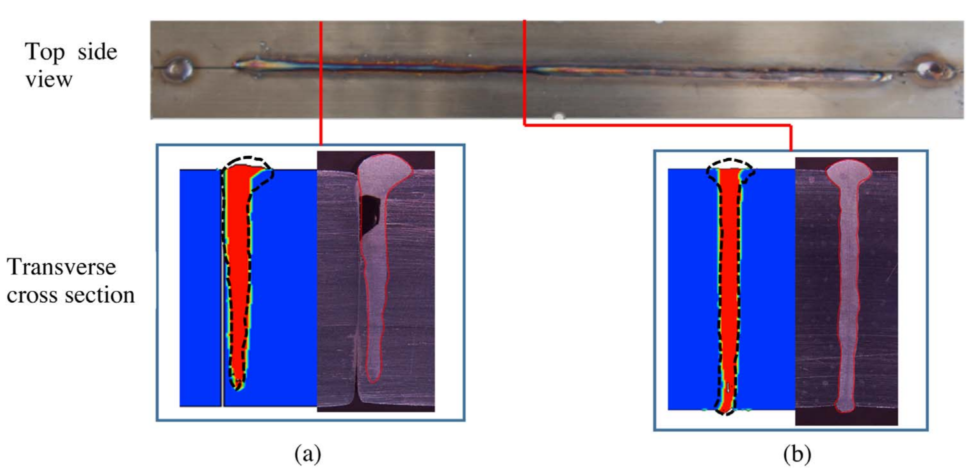

Fig. 6. Comparison of the experimental and simulation results. (a) Deviation between laser beam and gap: 0.5 mm; (b) Laser beam focusing on the gap.

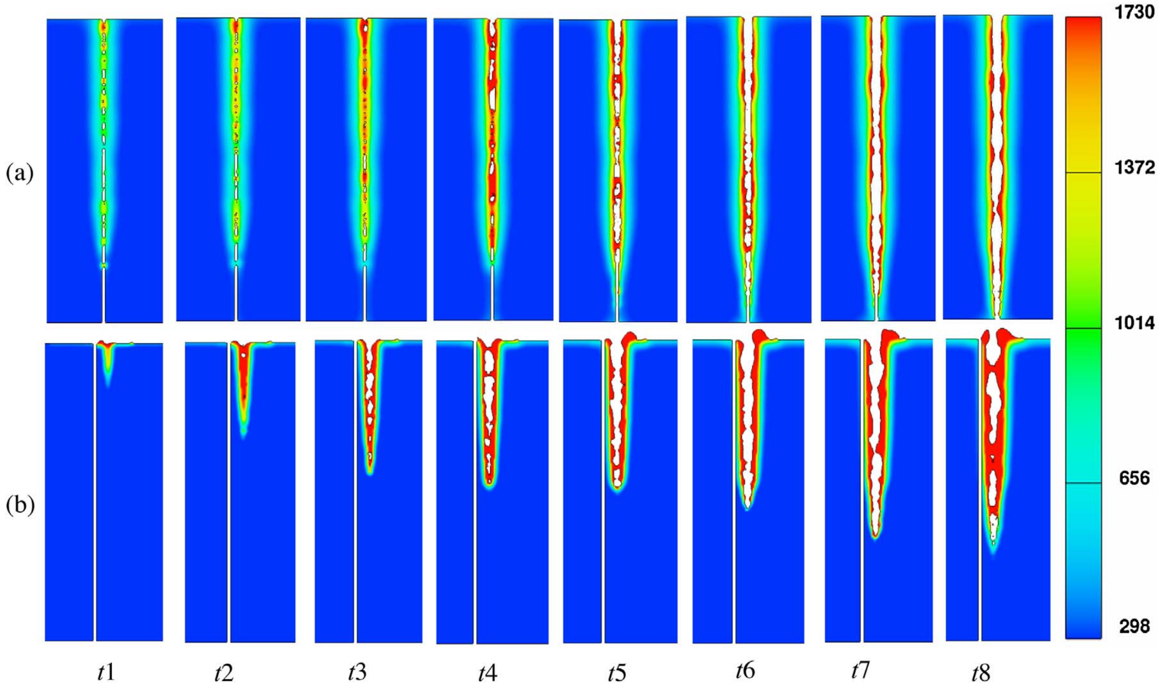

Fig. 7. Formation of the keyhole comparison. (a) Laser beam focusing on the gap; (b) Deviation between laser beam and gap: 1 mm.

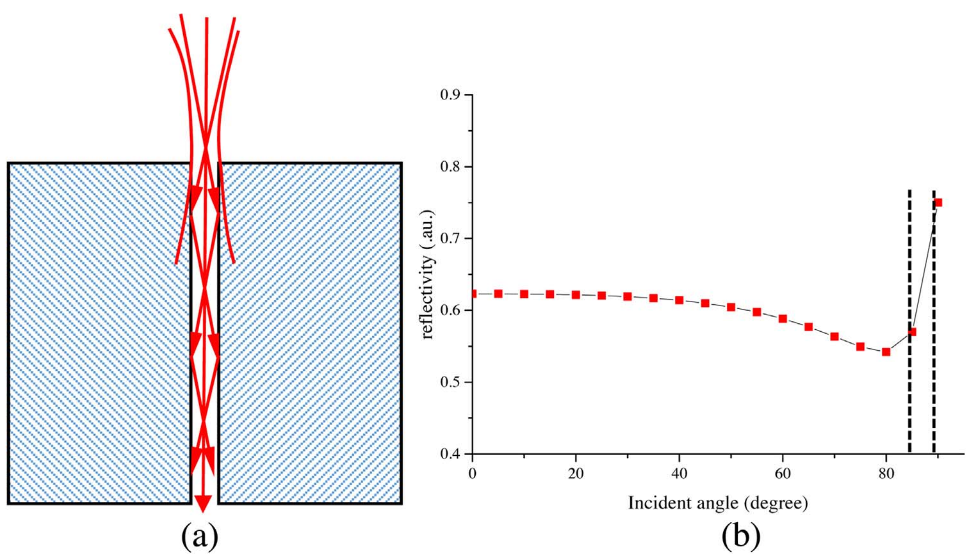

Fig. 8. The comparison of the formation of the keyhole. (a) Laser beam focusing on the gap; (b) Reflectivity curve depending on incident angle when $\varepsilon = 0.25$.

Especially, full penetration was achieved for Case 1 as shown in Fig. 9(a) when the keyhole depth was -10 mm from t = 40 ms to t = 60 ms, and the keyhole radius became smaller. The reason for this is that when the laser beam moved close to the gap center, the gap aided the transmission of the laser beam energy deep into the inner part of the material as discussed in section 4.1. The laser beam totally deviated from the gap and focused on the workpiece after t = 60 ms, and the keyhole radius increased while the keyhole depth decreased. Fig. 9(b) shows the keyhole radiuses and keyhole depths are similar to Case 1 when the laser beam focused on the workpiece. The keyhole radiuses and keyhole depths in Fig. 9(c) were similar with Fig. 9(b). When the laser beam focused on the gap shown in Fig. 9(d), once the full penetration was achieved, it would exist until the welding process finished. The average keyhole radius was similar with the average keyhole radius from t = 40 ms to t = 60 ms in Fig. 9(a).

4.2. Variation of absorbed energy by the workpiece

The variation of attenuation effect induced by the plume was studied by Greses et al., 2004. They found that the distribution of the plume can highly affect the attenuation effect. Tan (2013) calculated the maximum attenuation coefficient, which is about 22% of the total power input, by applying a three-dimensional simulation method. In this work, full penetration was achieved from t = 40 ms to t = 60 ms in Case 1 and the plume on the entrance of the keyhole would be reduced (this phenomenon will be discussed in detail in section 4.4). Therefore, the attenuation coefficient induced by the plume was set at 10% of the input laser power in this work.

The energy absorbed by the workpiece was calculated by tracing each laser ray. In Fig. 10(a), when the full penetration was produced during t = 40 ms to t = 60 ms, the absorbed energy decreased quickly because some laser rays could not contact the workpiece due to the gap or escaped through the gap. After that, laser beam transversally crossed

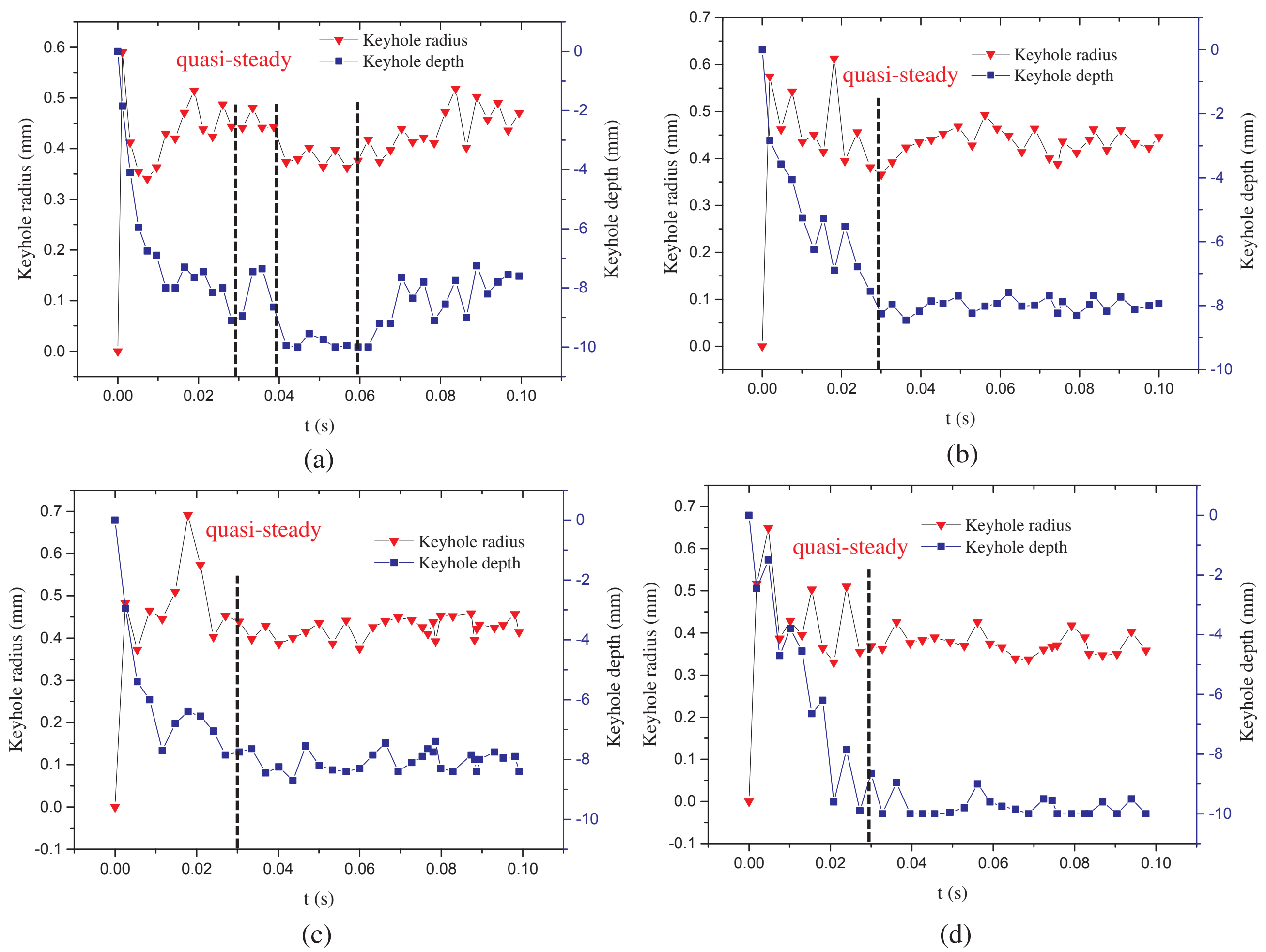

Fig. 9. Variation of keyhole radiuses and keyhole depths. (a) Case 1-Laser beam traveling cross the gap; (b) Case 2-Deviation between laser beam and the gap:1 mm; (c) Case 3-Deviation between laser beam and the gap:0.5 mm; (d) Case 4-Laser beam is focusing on the gap.

the gap and focused on the other workpiece, and the absorbed energy increased again. In Fig. 10(d), in the first 30 ms, no full penetration was observed even as the laser beam focused on the gap, but the absorbed energy had higher fluctuation compared with the other three cases in the same period. When full penetration appeared after about $t = 30$ ms, the absorbed energy by workpiece fluctuated severely, but was similar with the fluctuation when laser beam focused on the gap in Fig. 10(a) in the center period. Fig. 10(b) and (c) show the similar curve of absorbed energy by the workpiece with small fluctuation as the keyholes in these two cases are closed at the bottom.

4.3. Pressure on the free surface and molten pool velocity

The average and maximum pressure on the free surface in Case 1 were calculated and shown in Fig. 11. The average pressure became lower from $t = 40$ ms to $t = 60$ ms due to the decreased absorbed energy by the workpiece when the laser beam focused on the gap. In Fig. 11(b), the maximum pressure on the free surface increased from 0 to 35 k Pa in very short time at the start of the process, and then decreased to 10 k Pa at $t = 20$ ms. The maximum pressure on the free surface became steady as it was mainly caused by recoil pressure induced by evaporation, and the evaporation due to the absorbed laser energy mostly took place in local areas.

Two probes were employed to investigate the velocities of the molten pool. Each probe was set at the same position in the four different simulations. The velocities of these two probes were calculated from $t = 46$ ms to $t = 53$ ms as the full penetration was achieved in Case 1 at this period. The results are shown in Fig. 12, and the average velocities are tabulated in Table 3. The average velocities of P1 and P2 in Case 1 and Case 4 are similar when the full penetration was observed, but smaller than the velocities in Case 2 and Case 3. The reason for this could be that the absorbed energy of the laser beam by the workpiece decreased, and the total recoil pressures on the free surface decreased consequently when full penetration was achieved. Therefore, the momentum of the molten pool decreased.

4.4. Plume distribution and velocity calculation

You et al. (2013) have experimentally studied the distribution of the plume in partial penetration and full penetration laser welding. The material is 304 stainless steel, and the thickness of workpiece is 6 mm in their experiment. With laser power of 3 kW under the welding speed of 2 m/min, only partial penetration was achieved and the plume was ejected only through the top side, as shown in Fig. 13(a). When the laser power was increased to 4 kW, full penetration appeared and the plume was ejected out of both the top and bottom side of the workpiece, as shown in Fig. 13(b). Their studies also revealed that mean values of the keyhole size during 3 kW and 4 kW laser power welding which equaled 0.1438 mm^2 and 0.1305 mm^2, respectively, was slightly smaller than keyhole size obtained from the simulation in this work (The difference is due to the higher laser power of 6 kW which was used in this work). The mean intensity values of plume visible light were measured as 0.8662 and 0.7624 respectively, which represented the intensity or density of the plume. Fig. 13(a) and (b) show the different

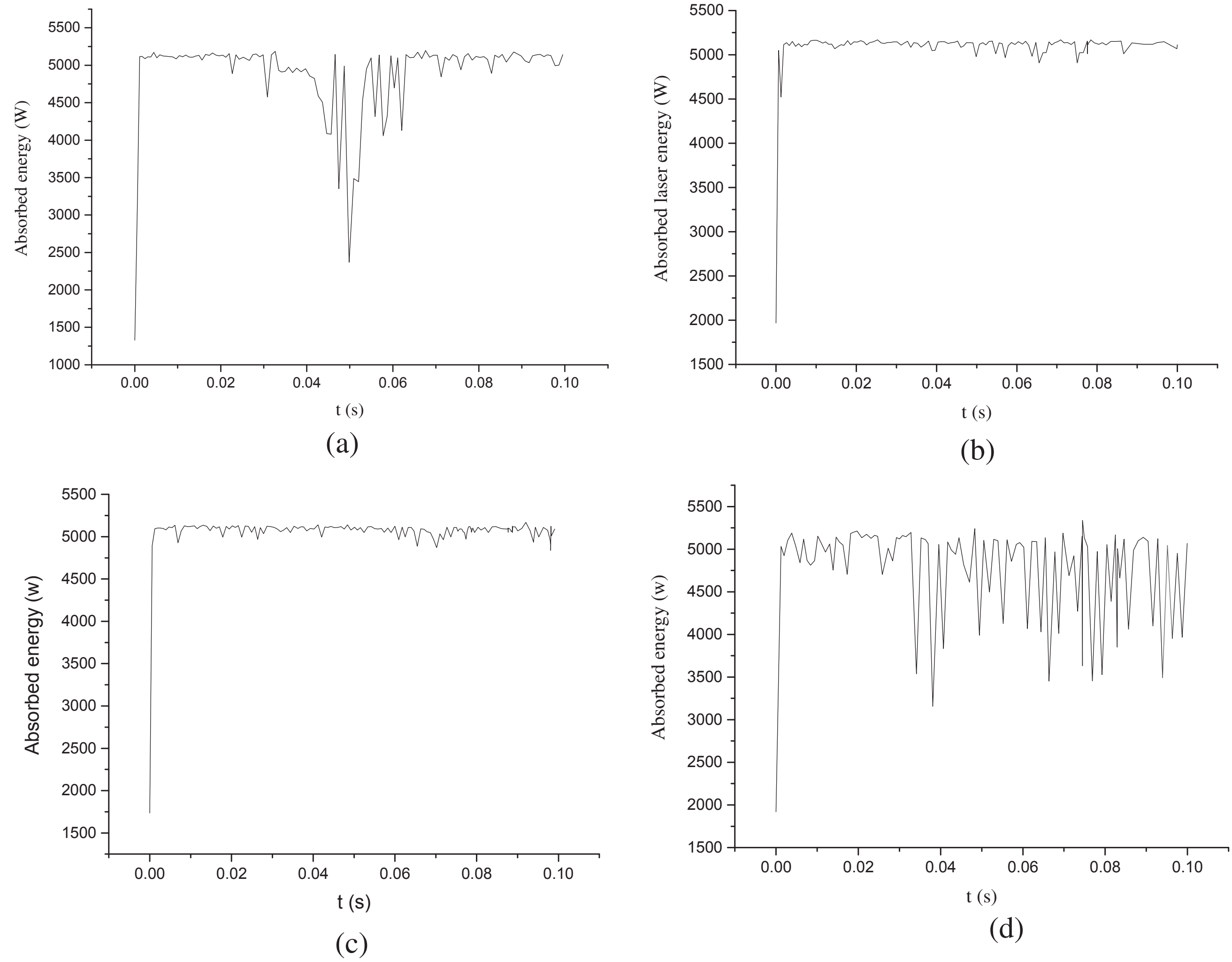

Fig. 10. Variation of absorbed energy in the four simulations. (a) The variation of absorbed energy in Case 1; (b) The variation of absorbed energy in Case 2; (c) The variation of absorbed energy in Case 3; (d) The variation of absorbed energy in Case 4.

distribution of plume on the top side of the workpiece. The smaller volume of the top plume in Fig. 13(b) reduced the attenuation effect of the laser beam energy and enhanced the absorbed energy by the workpiece. To explain the different distribution of the plume in Fig. 13, the velocity of the plume in Case 1 was calculated with the Cho's method (Cho et al., 2012). The result is shown in Fig. 14. When full penetration was achieved between $t = 40$ ms to $t = 60$ ms, the energy absorbed by workpiece decreased, consequently the total evaporation induced by the laser energy reduced and resulted in the lower velocities of the plume at the exits of the keyhole. Therefore, the momentum of

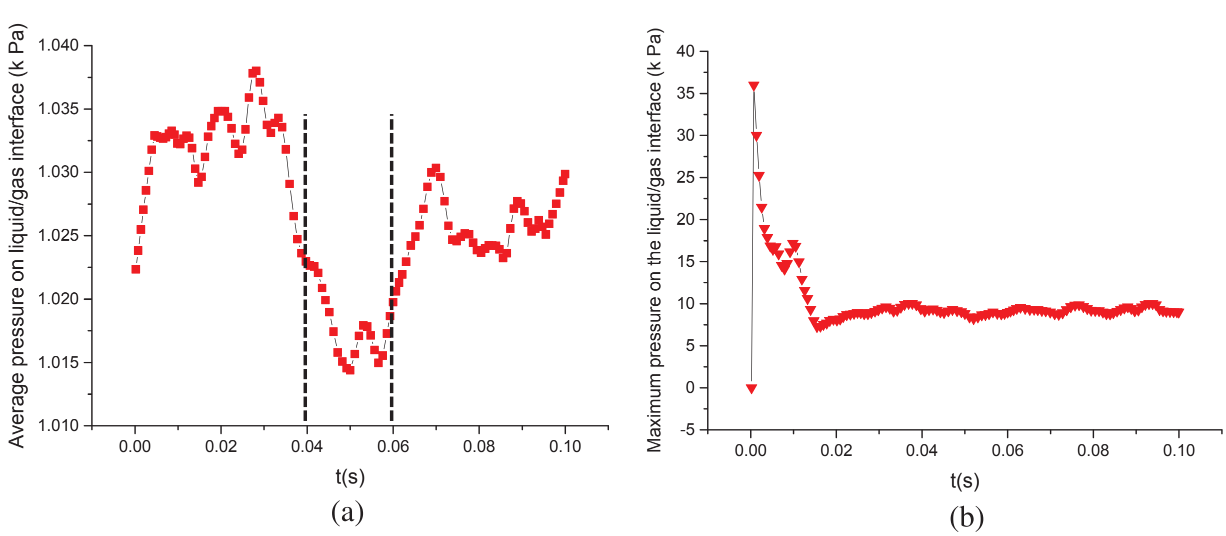

Fig. 11. Variation of average and maximum pressure on the free surface. (a)Variation of the average pressure; (b) Variation of the maximum pressure.

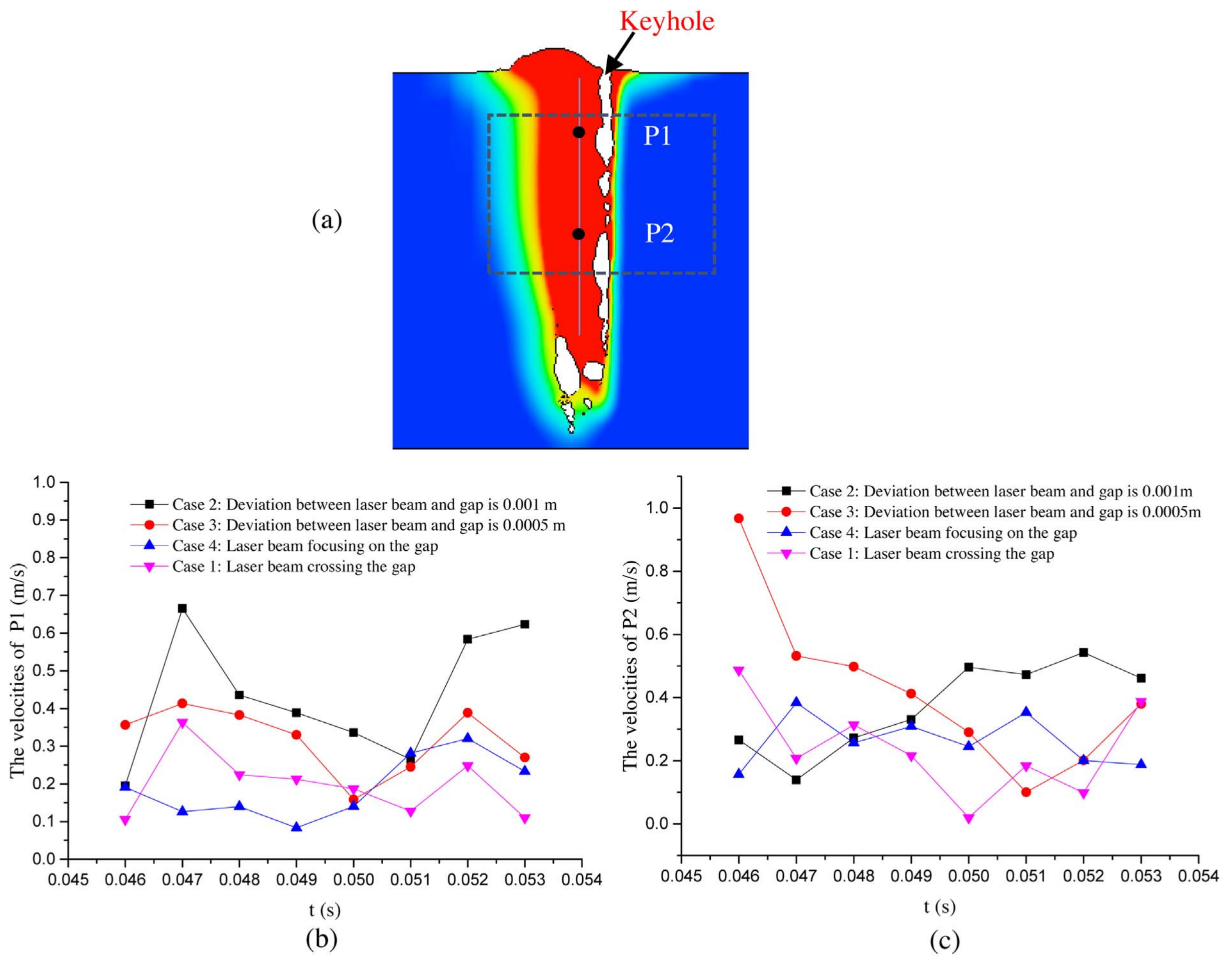

Fig. 12. Comparison of velocities of the two probes in molten pool. (a) Schematic of location of P1 and P2; (b) Velocities of P1 in four different Cases; (c) Velocities of P2 in four different Cases.

Table 3
Comparison of average velocity of P1 and P2 in the four cases.

Probe (m/s)	Case 1	Case 2	Case 3	Case 4
Average velocity of P1	0.197166	0.436635	0.318143	0.189452
Average velocity of P2	0.239231	0.372701	0.422767	0.261595

the plume became less than in the other periods. Meanwhile, the plume was ejected out through both the top and bottom exits. This could be an explanation of the decrease in light intensity and plume volume when processing with a laser power of 4 kW (Fig. 13(b)) compared to a processing with a laser power of 3 kW (Fig. 13(a)).

4.5. Flow pattern in the molten pool

The flow patterns of the molten pool in Case 2, 3 and 4 at t = 49 ms, 50 ms, 51 ms and 52 m s were investigated and shown in Fig. 15. The streamlines of the longitudinal velocity field were also calculated. The flow types of Case 2 and Case 3 are almost the same. The flows marked as “s” for Case 2 and Case 3 were mainly caused by the Marangoni force due to the high gradient of temperature between the surrounding area and central area of the molten pool. The other flows were induced by a combination of the recoil pressure, gravity, surface tension, and buoyancy force. The flow patterns in Case 4 were entirely different with that of Case 2 and 3, and were more complex and irregular. This can be

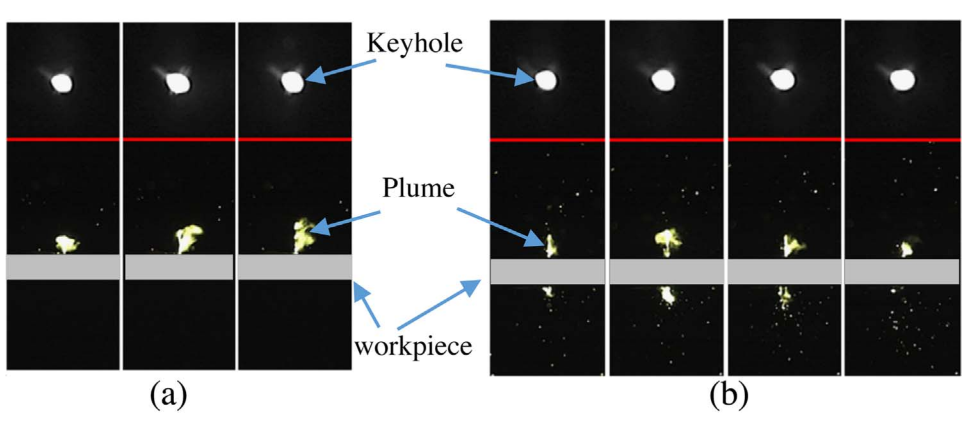

Fig. 13. The distribution of the plume and the keyhole areas variation in reference (You et al., 2013). (a) Laser welding with partial penetration;(b) Laser welding with full penetration.

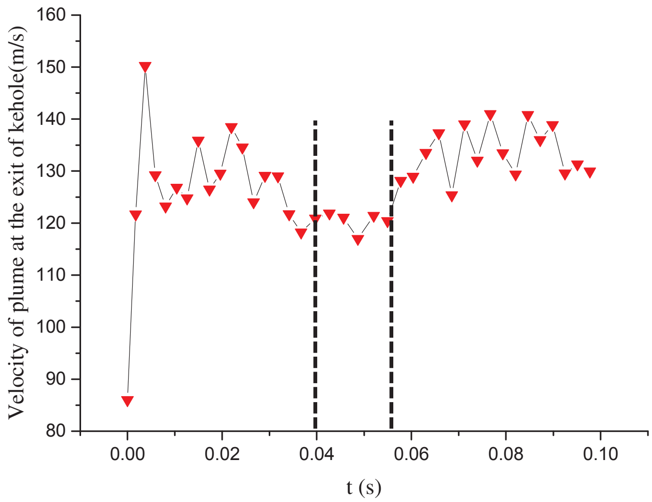

Fig. 14. The variation of the plume velocity in Case 1.

explained as follows. In Case 2 and 3, the keyhole was closed at the bottom due to partial penetration, keeping the beam absorbed energy at a stable, as shown in Fig. 10 (b) and (c). Also, a quasi-equilibrium state of the molten pool was achieved. However, in Case 4 the keyhole was open at the bottom and laser rays escaped through the gap, with absorbed energy highly fluctuating as some laser rays escaped through the gap (see Fig. 10(d)). The high fluctuation of absorbed energy resulted in the direct variation of the recoil pressure. Therefore the quasi-equilibrium of the molten pool was difficult to be achieved, and the flow pattern in Case 4 became complex and irregular.

5. Conclusions

In this work, a three-dimensional numerical model was established to investigate the effect of the joint gap during laser butt welding. The energy transmission of the laser beam, the dynamics of the keyhole, molten pool and plume in laser butt welding were also discussed. The results of this work can be summarized as follows:

(1) The gap in laser butt welding contributes to transmitting the energy of laser beam deep into the inner part of the workpiece.
(2) With the same laser power of 6 kW, full penetration and narrow fusion zone were achieved when the laser beam was focused on the gap; partial penetration and wide fusion zone were obtained when the laser beam deviated from the gap and was focused on the workpiece.
(3) The absorbed energy by the workpiece decreased when the laser beam was focused on the gap as some rays escaped through the gap. The decrease in absorbed energy resulted in a lower velocity of the molten pool and lower average pressure on the free surface. The velocity and volume of the plume on the top side of the workpiece also decreased.
(4) The flow patterns in laser butt welding were more complex and irregular than it in bead on plate laser welding due to the high fluctuation of the energy absorbed by the workpiece.

The influences of different gap sizes, different inclined angles of the

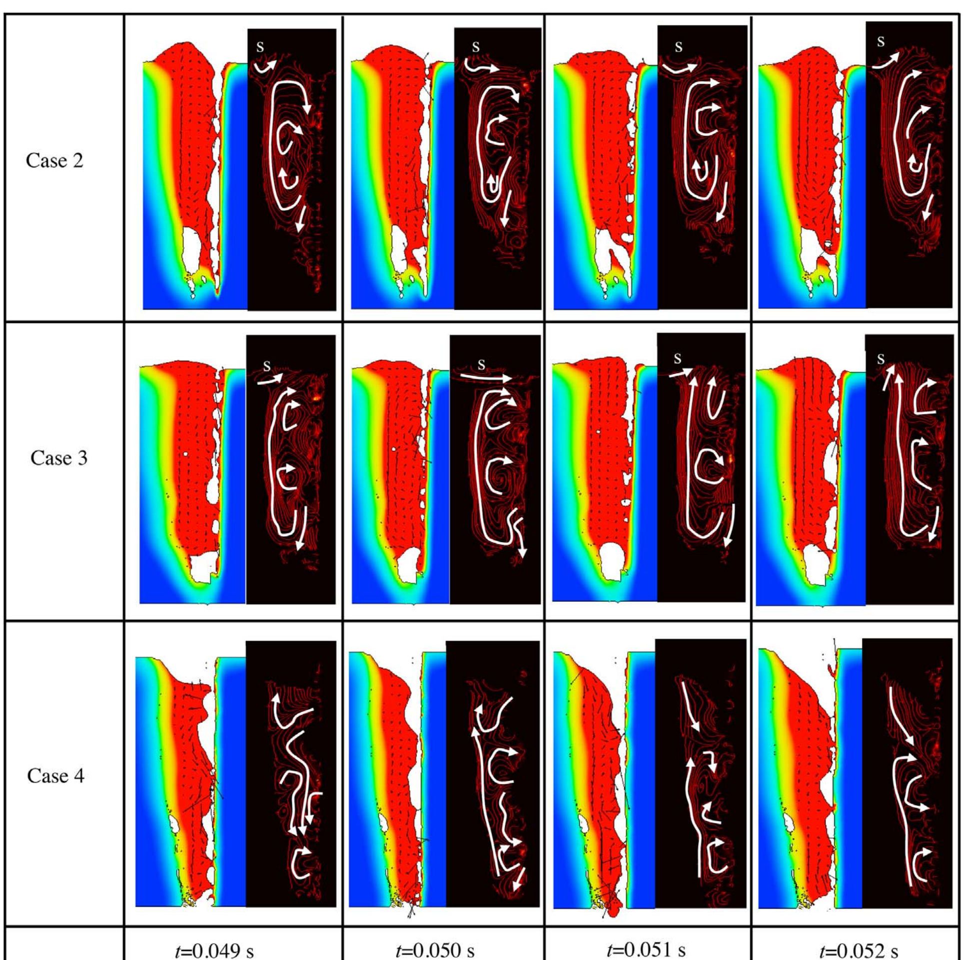

Fig. 15. Flow patterns and calculated streamlines in Case 2, 3 and 4.

laser beam, different powers and diameters of the laser beam, different focal positions and different welding speeds during laser butt welding will be investigated in the authors' future work.

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References

Ahn, J., Na, S.J., 2013. Three-dimensional thermal simulation of nanosecond laser ablation for semitransparent material. Appl. Surf. Sci. 283, 115–127.
Allmen, M., Blatter, A., 1995. Laser–Beam Interactions with Materials, second ed. Springer, New York.
Amara, E.H., Fabbro, R., 2008. Modelling of gas jet effect on the melt pool movements during deep penetration laser welding. J. Phys. D: Appl. Phys. 41, 055503.
Amara, E.H., Fabbro, R., 2010. Modeling of humps formation during deep-penetration laser welding. Appl. Phys. A 101, 111–116.
Atabaki, M.M., Yazdian, N., Ma, J., Kovacevic, R., 2016. High power laser welding of thick steel plates in a horizontal butt joint configuration. Opt. Laser Technol. 83, 1–12.
Chen, L., Zhou, L., Tang, C., Huang, W., Wang, C., Hu, X., Shao, X., 2014a. Study of laser butt welding of SUS301L stainless steel and welding joint analysis. Int. J. Adv. Manuf. Technol. 73, 1695–1704.
Chen, S., Zhang, M., Huang, J., Cui, C., Zhang, H., Zhao, X., 2014b. Microstructures and mechanical property of laser butt welding of titanium alloy to stainless steel. Mater. Des. 53, 504–511.
Cho, J.H., Na, S.J., 2006. Implementation of real-time multiple reflection and Fresnel absorption of laser beam in keyhole. J. Phys. D: Appl. Phys. 39, 5372–5378.
Cho, W.I., Na, S.J., Cho, M.H., Lee, J.S., 2010. Numerical study of alloying element distribution in CO2laser/GMA hybrid welding. Comput. Mater. Sci. 49, 792–800.
Cho, W.I., Na, S.J., Thomy, C., Vollertsen, F., 2012. Numerical simulation of molten pool dynamics in high power disk laser welding. J. Mater. Process. Technol. 212, 262–275.
Dal, M., Fabbro, R., 2016. [INVITED] An overview of the state of art in laser welding simulation. Opt. Laser Technol. 78, 2–14.
Gao, X.D., Sun, Y., 2014. Monitoring of high power disk laser welding of Type 304 austenitic stainless steel based on keyhole dynamic characteristics. Insight Non-Destruct. Test. Cond. Monit. 56, 312–317.
Gou, N.N., Zhang, J.X., Zhang, L.J., Li, Z.G., Bi, Z.Y., 2016. Single pass fiber laser butt welding of explosively welded 2205/X65 bimetallic sheets and study on the properties of the welded joint. Int. J. Adv. Manuf. Technol. 86, 2539–2549.
Greses, J., Hilton, P.A., Barlow, P.A., Steen, W.M., 2004. Plume attenuation under high power Nd: yttritium-aluminum-garnet laser welding. J. Laser Appl. 16, 9–15.
Hu, C.Y., Zhang, F.J., 2001. Narrow gap welding technology and economical characters analysis. Weld. Technol. 30, 47–54.
Kawahito, Y., Matsumoto, N., Abe, Y., Katayama, S., 2011. Relationship of laser absorption to keyhole behavior in high power fiber laser welding of stainless steel and aluminum alloy. J. Mater. Process. Technol. 211, 1563–1568.
Lee, H.T., Chen, C.T., Wu, J.L., 2010. Numerical and experimental investigation into effect of temperature field on sensitization of Alloy 690 butt welds fabricated by gas tungsten arc welding and laser beam welding. J. Mater. Process. Technol. 210, 1636–1645.
Li, R., Wang, T., Wang, C., Yan, F., Shao, X., Hu, X., Li, J., 2014. A study of narrow gap laser welding for thick plates using the multi-layer and multi-pass method. Opt. Laser Technol. 64, 172–183.
Rai, R., Palmer, T.A., Elmer, J.W., Debroy, T., 2009. Heat transfer and fluid flow during electron beam welding of 304L stainless steel alloy. Welding J. 88 (3), 54s.
Sudnik, W., Radaj, D., Breitschwerdt, S., Erofeew, W., 2000. Numerical simulation of weld pool geometry in laser beam welding. J. Phys. D: Appl. Phys. 33, 662–671.
Tan, W.D., Bailey, N.S., Shin, Y.C., 2013. Investigation of keyhole plume and molten pool based on a three-dimensional dynamic model with sharp interface formulation. J. Phys. D: Appl. Phys. 46, 055501.
Wu, C.S., Wang, H.L., Zhang, Y.M., 2009. Numerical analysis of the temperature pro-files and weld dimension in high power direct-diode laser welding. Comput. Mater. Sci. 46, 49–56.
You, D.Y., Gao, X.D., Katayama, S., 2013. Multiple-optics sensing of high-brightness disk laser welding process. Ndt E Int. 60, 32–39.
Zhang, Y.X., Gao, X.D., Katayama, S., 2015. Weld appearance prediction with BP neural network improved by genetic algorithm during disk laser welding. J. Manuf. Syst. 34, 53–59.
Zhao, H., DebRoy, T., 2003. Macroporosity free aluminum alloy weldments through numerical simulation of keyhole mode laser welding. J. Appl. Phys. 93, 10089–10096.
Zhao, H.Y., Niu, W.C., Zhang, B., Lei, Y.P., Kodama, M., Ishide, T., 2011. Modelling of keyhole dynamics and porosity formation considering the adaptive keyhole shape and three-phase coupling during deep-penetration laser welding. J. Phys. D: Appl. Phys. 44, 485302.
Zhou, J., Tsai, H.L., Lehnhoff, T.F., 2006. Investigation of transport phenomena and defect formation in pulsed laser keyhole welding of zinc-coated steels. J. Phys. D: Appl. Phys. 39, 5338.